Worksheet — Normal Force Simulation

Use with *Simulation - Normal force.html* · goes with carousel Station 1 (weight and normal force)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/02-normal-force** in a browser (offline: *Simulation - Normal force.html*). Choose the situation at the bottom: **Push / pull**, **Ramp**, **Lift** or **Stack**. The **Tasks** tab has the same three parts as this sheet. The simulation checks your numbers; write your **explanations** here.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab, Station 1 (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use g = 9.81 N kg^{-1}. A scale shows F_N/g in kg.

Scan to open

Part 1 — Pre-lab: predict, then test

Write your prediction **before** you test it. Lock it in on the sim, test it, then explain.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. You press down on a 1.00 kg mass on a kitchen scale with 5.0 N. What does the scale read?			
2. You pull up on the same mass with 5.0 N (it stays on the scale). What does the scale read?			
3. The scale with the 1.00 kg mass is tilted to 30°. What happens to the reading?			
4. A 60 kg student is on a bathroom scale in a lift that speeds up going up. Reading?			
5. The lift cable snaps and the lift falls freely. Reading?			
6. A 2.0 kg block sits on a 3.0 kg block on a table. How hard does the table push up?			

7. Draw the free-body diagram of the 1.00 kg mass while you push down with 5.0 N. Then calculate F_N (five steps).

Free-body diagram and working

F_N = _____ N Scale reading = _____ kg

Part 2 — During the lab: Station 1

1.00 kg mass on the kitchen scale. Use the sim, then compare with your Station 1 measurements.

Angle θ / °	Sim reading / kg	Predicted m cos θ / kg	Our reading / kg	$F_N = mg$ cos θ / N
0				
15				
30				

8. Give **one** reason why your measured readings differ from the simulation.

9. Why does the scale read less on the slope? Which force holds the mass up **along** the slope?

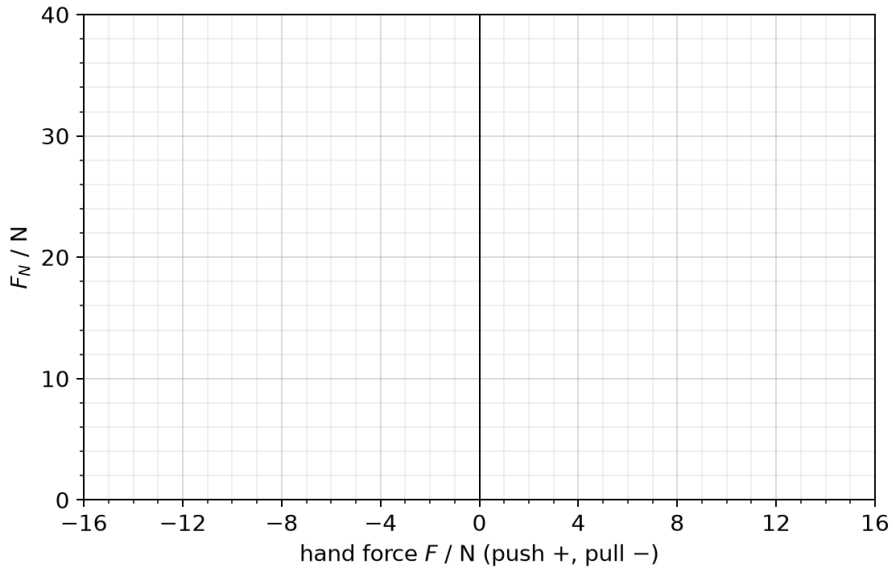
10. At Station 1 you pressed down with one finger. Explain what happened to the reading using the FBD.

Part 3 — Homework: analyse

A. Pushing and pulling (m = 2.00 kg)

Hand force F / N (push +)	-15	-10	-5	0	5	10	15
F_N / N							

11. Plot F_N against F and draw the best-fit line.

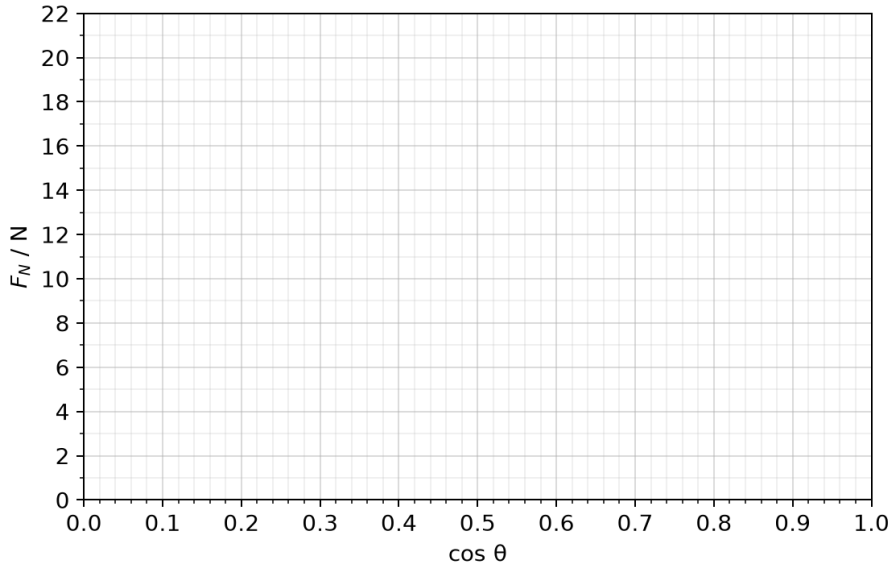


12. Gradient = _____ Intercept = _____ N. What does each one mean physically?

B. Tilting (m = 2.00 kg)

θ / °	0	10	20	30	40
cos θ					
F_N / N					

13. Plot F_N against cos θ. Gradient = _____ N. Which quantity does the gradient equal?



C. Lift (Newton's 2nd law — Blocks 3–4)

14. A 60 kg student stands on a bathroom scale in a lift that accelerates **upward** at 1.5 m s^{-2}. Draw the FBD and calculate the scale reading in N and in kg.

Free-body diagram and working

15. The lift now **slows down while going up** (acceleration 1.5 m s^{-2} downward). What does the scale show in kg?

Working

16. Explain why the scale reads 0 kg in free fall, even though the student's weight is still 589 N.

D. Stack

17. Blocks 1 (3.0 kg, bottom), 2 (2.0 kg) and 3 (1.0 kg, top) are stacked on a table. Draw the FBD of block 1 and calculate the force of block 2 on block 1 and the force of the table on block 1.

Free-body diagram and working

18. Name the Newton's 3rd-law partner of 'the force of block 2 on block 1'. Why is the weight of block 2 **not** its partner?